

1.3.2) To estimate the actual power received (P_{rx}) at a given location (x, y) on the campus map, the power density is integrated over the effective aperture of the receiver antenna (A_e):

$$P_{rx}(x, y) = S_{avg}(x, y) \cdot A_e \quad (\text{eq.13})$$

1.3.2) Finally, the received power is converted to decibel-milliwatts (dBm), providing a standard scale for signal strength measurement and heat map visualization:

$$P_{dBm} = 10 \cdot \log_{10} \left(\frac{P_{rx}}{1 \text{ mW}} \right) \quad (\text{eq.14})$$

2. Node Placement

The Placement process begins by transforming the map into a digital grid where the pixel intensity represents signal viability.

Target Points Matrix (D): This matrix identifies dead zones (buildings or concrete blocks) requiring signal coverage. It is constructed by extracting pixels where their brightness intensity is below the threshold of 85 (**Pixel < 85**), mapping them as the primary targets for coverage.

Candidate Locations Matrix (L): This matrix defines the feasible node places; it includes all potential coordinates where a node can be physically placed. Only pixels with a brightness intensity above 140 (**Pixel > 140**) are selected to ensure nodes are placed in small corridors and away from buildings.

2.1) Euclidean Distance Modelling for Node Coverage

2.1.1) To ensure spatial accuracy, we utilize the **Euclidean Distance formula** to determine the relationship between nodes and their coverage areas, defined as follows:

$$d_{i,j} = \sqrt{((x_i - x_j)^2 + (y_i - y_j)^2)} \quad (\text{eq.15})$$

2.1.2) To minimize the interference near the signal source, we enforced a no-placement zone within a radius R from the source. Any candidate node is discarded if $d_{\text{source,node}} \leq R$.

2.1.3) Optimal Radius Selection (R): An iterative search algorithm determines the optimal R that balances the safety exclusion zone with the global node cost, expressed as:

$$R_{opt} = \arg \max_R \{R \mid f(R) = \min(\text{Nodes})\}$$

2.2) Coverage Adjacency Matrix (A):

The Adjacency Matrix $A_{N \times M}$ serves as the core mapping component, where N is the number of dead points and M is the number of candidate locations: $A_{i,j} = 1$: If the candidate node j successfully covers the target pixel i (i.e., $d_{(i,j)} < 50$ pixels). $A_{i,j} = 0$: If the target pixel i is outside the node's reach.

2.3) Euclidean Distance Modelling for Node Coverage

Integer Linear Programming (ILP) Formulation: The final node placement is solved using the `intlinprog` algorithm, which optimizes the deployment based on the following constraints:

2.3.1) Objective Function (Cost Minimization), given by:

$$\min Z = \sum_{j=1}^N x_j, \quad x_j \in \{0, 1\} \quad (\text{eq.16})$$

Where x_j is a binary decision variable indicating whether a node is deployed (**1**) or not (**0**), Z is the total number of deployed nodes.

2.3.2) Coverage Constraint (Set Cover Problem): Ensures that every pixel in the D matrix is covered by at least one active node, given by:

$$\sum_{j=1}^N A_{i,j} x_j \geq 1, \quad \forall i \in \text{Dead Zones} \quad (\text{eq.17})$$

3. Node Optimization

This technical specification details the mathematical framework designed to optimize the deployment and spatial configuration of Wi-Fi nodes. The model executes a multi-stage pipeline, transitioning from raw visual data to a refined spatial topology that maximizes coverage while strictly mitigating co-channel interference.

3.1) Image Pre-processing: Gray-Level Transformation

The optimization begins with the dimensionality reduction of the input signal. The raw image data, represented as a three-dimensional array $f(x, y, \mathbf{3})$, must be converted into a single-dimension grayscale array $g(x, y)$. This step is critical to minimize the computational overhead required for the subsequent Circular Hough Transform (CHT).

To quantify this transformation, we apply the "Perceptual Weighting" formula to the Red (R), Green (G), and Blue (B) channels:

$$g(x, y) = wr \cdot R(x, y) + wg \cdot G(x, y) + wb \cdot B(x, y) \quad (\text{eq.18})$$

Technical Specification: ITU-R BT.601 Standard To ensure digital standardization and luminance accuracy, the model utilizes the following weighting coefficients:

wr (Red): 0.2989

wg (Green): 0.5870

wb (Blue): 0.1140

Executing this transformation yields the finalized grayscale equation:

$$g(x, y) = 0.2989R + 0.5870G + 0.1140B \quad (\text{eq.19})$$